

that of planet Neptune,
the graph for Uranus is one with smaller gradient [B1] with the same zero-y-intercept [B1].

3 (a)

Any 2 assumptions [B1x2]

- Large number of particles.
- The time of collisions is negligible compared to the time between the collisions.
- Negligible intermolecular forces (except during collisions)
- Total volume of the gas particles are negligible compared to volume of the containing vessel.
- Average separation of gas particles is very large compared to the size of particle. Negligible microscopic PE.
- The gas particles moves randomly in all directions.

3 (b)(i)

Mass of 1 mole of helium-4 gas = 4.0×10^{-3} kg

$$\begin{aligned}\text{Mass of 1 atom} &= 4.0 \times 10^{-3} / 6.02 \times 10^{23} \\ &= 6.6 \times 10^{-27} \text{ kg} \quad [A1]\end{aligned}$$

Alternatively

$$\begin{aligned}\text{Mass of one atom} &= 4u = 4 \times (1.66 \times 10^{-27}) \\ &= 6.6 \times 10^{-27} \text{ kg}\end{aligned}$$

Mass the 2 electrons is negligible and so can be ignored.

No marks if 4xmass of proton. He-4 has 2 protons and 2 neutrons

3 (b)(ii)

$$\begin{aligned}\text{Internal Energy } U &= \frac{3}{2} nRT = \frac{3}{2} (1.2) (8.31) (273 + 27) \\ &= 4.5 \times 10^3 \text{ J} \quad [A1]\end{aligned}$$

$$\text{Alternatively, can use } \frac{3}{2} NkT = \frac{3}{2} (nN_a) kT$$

3 (b)(iii)

$$\frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} kT \quad [C1]$$

$$\frac{1}{2} (6.6 \times 10^{-27}) \langle c^2 \rangle = \frac{3}{2} (1.38 \times 10^{-23}) (300) \quad [C1]$$

$$r.m.s. \text{ speed} = 1.4 \times 10^3 \text{ m s}^{-1} \quad [A0]$$

Alternatively,

Mean KE = Internal E / (no. of atoms)

$$\frac{1}{2} m \langle c^2 \rangle = \frac{4.49 \times 10^3}{6.023 \times 10^{23}} = 7.45 \times 10^{-21} \quad [C1]$$

$$\frac{1}{2} (6.64 \times 10^{-27}) \langle c^2 \rangle = \frac{4.49 \times 10^3}{6.023 \times 10^{23}} = 7.45 \times 10^{-21} \quad [C1]$$

$$r.m.s. \text{ speed} = 1.5 \times 10^3 \text{ m s}^{-1} \quad [A0]$$

3 (b)(iv)

The r.m.s. speed is a **singular representation of the effective speed** that determines the internal energy of the gas. It is a sort of an 'average' speed.

(MUST put inverted comma for the term 'average' as it denotes a proxy value and not the actual average speed. FYI, the average velocity is zero as the motion is random, Average speed is not the same as rms speed as the distribution of the speed does not follow a normal distribution).

Statistically, that means there are some particles that have speeds greater (or less) than the r.m.s. speed. Thus, some helium particles that have **with speed greater than the escape speed will escape** the Earth's atmosphere. [B1]